Gruebler–Kutzbach Mobility Formula for Planar Mechanisms

$$M = 3(n - 1) - 2\,j1 - j2$$

n = number of links

Include ground link (link 1)

Each free link = 3 DOF

j1 = full joints

Revolute (R), Prismatic (P)

Each removes 2 DOF

j2 = half joints

Cam pair, Gear pair

Each removes 1 DOF

Compound Joint Rule

If m links share one joint location –> count it as (m − 1) full joints

Example: 3 links at one pin = 2 full joints (j1 += 2, not 1)